Practical 1: Georeferencing a Toposheet in ArcMap

Aim

To georefrence a scanned toposheet by assigning real-world coordinates so it can be used for further GIS analysis in ArcMap.

Objectives

By the end of this practical, the we will be able to:

1. Understand the concept and need for georeferencing.
2. Load a scanned toposheet into ArcMap.
3. Add Ground Control Points (GCPs) using map grid intersections.
4. Perform transformation and adjust RMS error.
5. Save the georeferenced raster image.

Materials Required

- ArcMap (ArcGIS Desktop)
- Scanned toposheet (JPG/PNG/TIFF)
- Knowledge of the coordinate system (e.g., UTM/Lat-Long)
- Grid values printed on the toposheet

Introduction

What is Georeferencing?

Georeferencing is the process of assigning spatial (real-world) coordinates to an image so it aligns correctly with other GIS data layers.

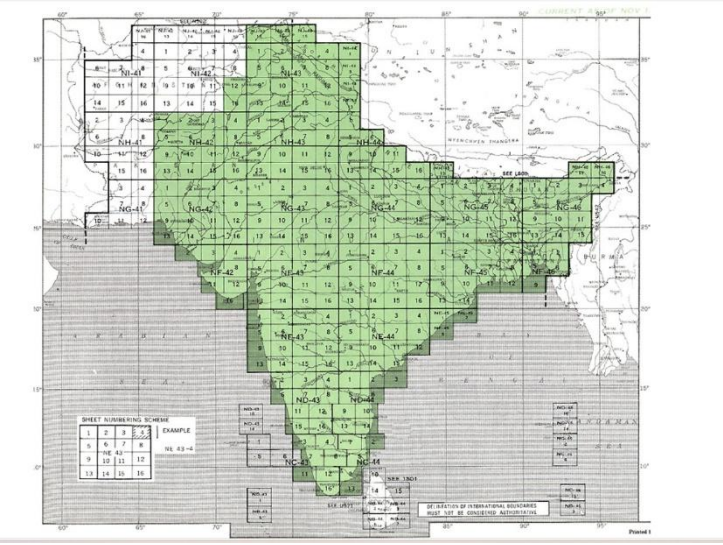
Why is Georeferencing Needed?

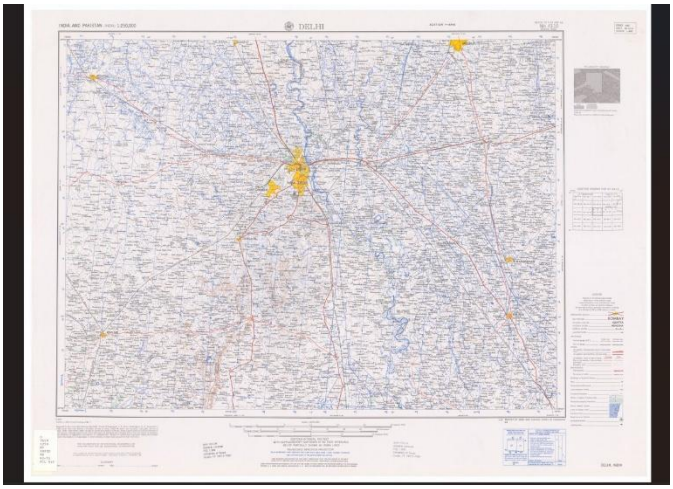
- To extract spatial features

- To digitize roads, rivers, contour lines
- To overlay the map with GIS datasets
- To integrate it into a GIS environment

Control Points

Control Points are known coordinate locations on the toposheet grid (e.g., longitude–latitude or easting–northing intersections)





Methodology

Step 1:- First we go to google site and download the toposheet of India and Pakistan (select the Surat map and save this picture on the folder).

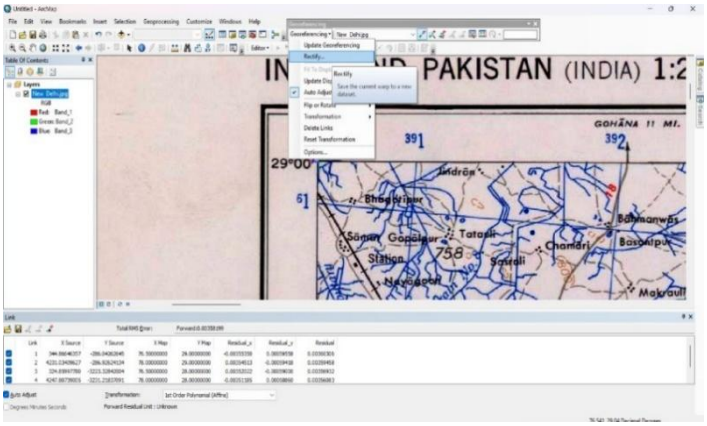
Step 2: Open the Arcgis application followed by new blank map.

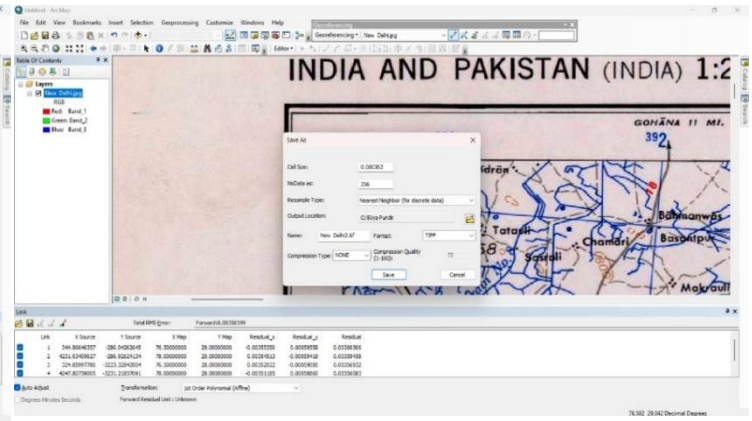
Step 3: Click on the toolbar and select the georeferencing toolbar list apply on the screen.

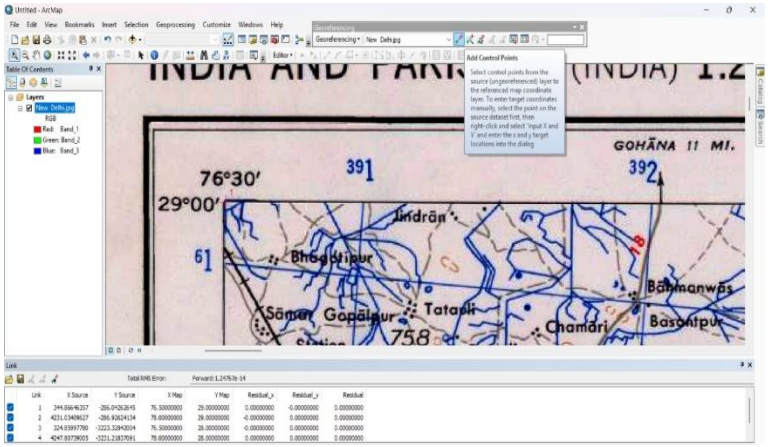
After then we add data of save the Surat map picture on folder represent as a (New Delhi.jpg). Apply on the layer box and image display on screen

Step 4: Then clicking on the add control points, go to top corner side then right click showing the input X and Y coordinates. Put the X and Y value in all four side. Then open the view link table showing all the 4 coordinates value of X and Y and we generate value on the map.

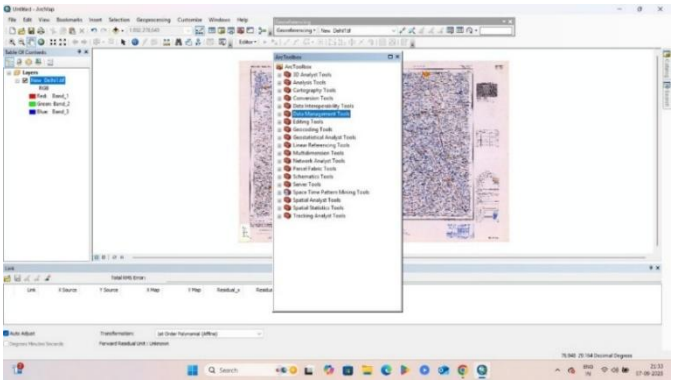
Step 5: Select the rectify option; then select the output location where we save TIFF image of new Surat in which folder and save it.





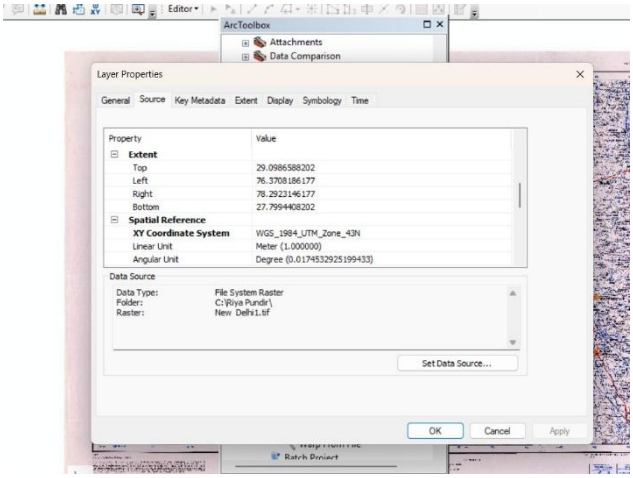


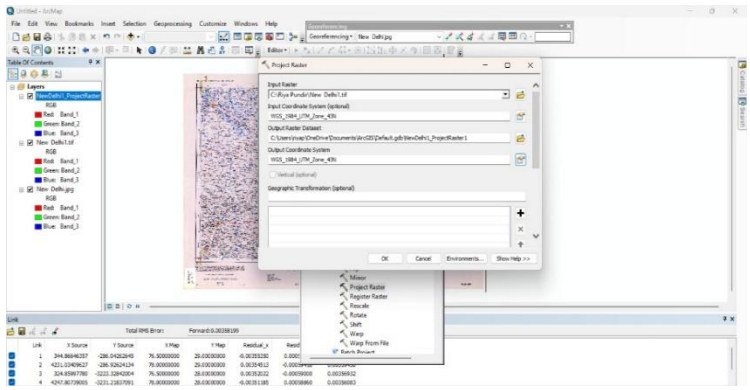
STEP 6: Go to layers right click then select properties; display should be in degrees decimal second; then apply and ok.

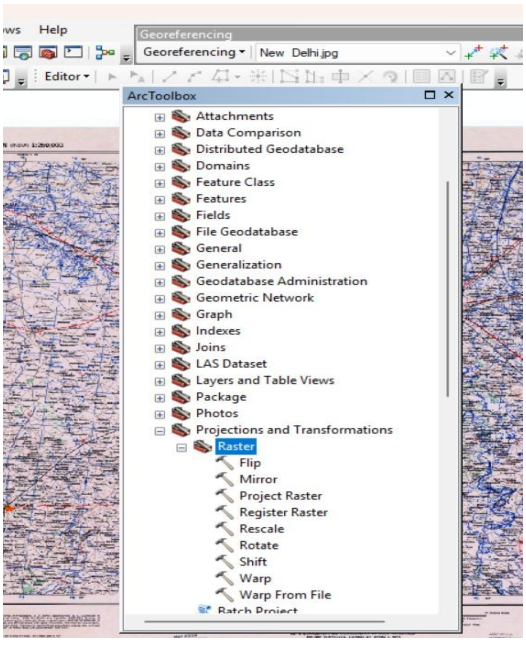


STEP 7: Click the Arc Toolbox; then go to Data Management Tools; click Projections and transformation; click the raster and then click on the project raster

STEP 8 : Select the input raster where we should save the file of the projection vision (c;\deepika\ new tiff) and go to output coordinate system where we give a (WGS_1984_UTM_Zone_ 43N) and input coordinates system optional (WGS_1984_UTM_Zone_43N) is not necessary to give information and apply ok .







Output

A properly georeferenced toposheet aligned with the coordinate system and ready for digitization or spatial analysis.

Precautions

- Choose widely spread control points (avoid clustering).
- Enter correct coordinates from the toposheet grid.
- Check that the correct coordinate system is selected.
- Avoid blurred map edges.

Conclusion

This exercise work has oriented to evaluate the accuracy that can be obtained by adjusting map image using polynomial equation with the help of longitude and latitude. Moreover, to compare between results that can be obtained applying first, second and third orders of polynomial in georefrencing of image further gets used for the purpose of digitization (point, line and polygon map) final layout on map image

Practical 2: Digitization of Point, Line & Polygon Features in ArcMap

Aim

Digitization of features means converting physical or analog information—such as details from maps or toposheets into digital form so it can be stored, processed, and analyzed on a computer.

Objectives

Accuracy and Precision: Digital data is more accurate and easier to edit compared to traditional analog methods.

Storage and Retrieval: Digital formats allow large amounts of information to be stored safely and retrieved quickly whenever needed.

Analysis and Interpretation: Once features are digitized, various software tools can analyze them to identify patterns, make predictions, and support decision-making.

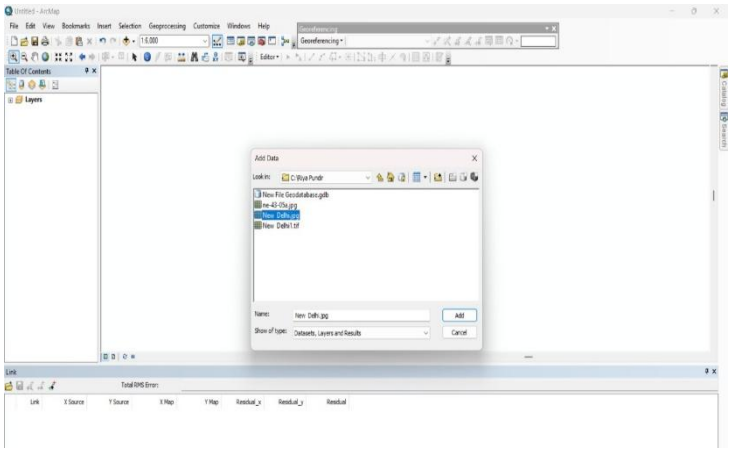
Interconnectivity: Digitized data can be shared and integrated across different systems, improving collaboration and overall understanding.

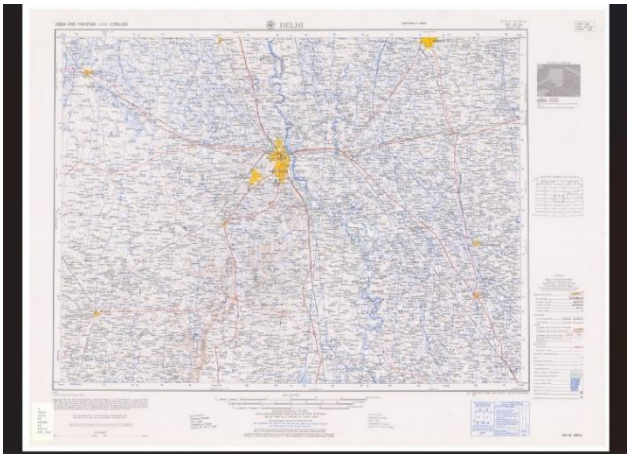
Data Used

A toposheet of Gujarat covering the Surat area was used. It was downloaded from the Educational Survey of India, toposheet number G4357 (45F/18), with a scale of 1:250000

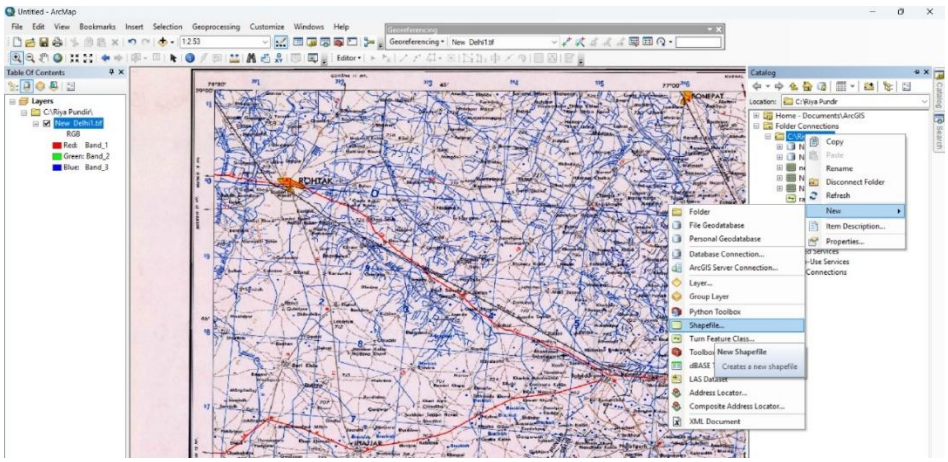
Methodology

Step 1. **Open ArcGIS:** Start ArcGIS Pro or ArcMap on your system.



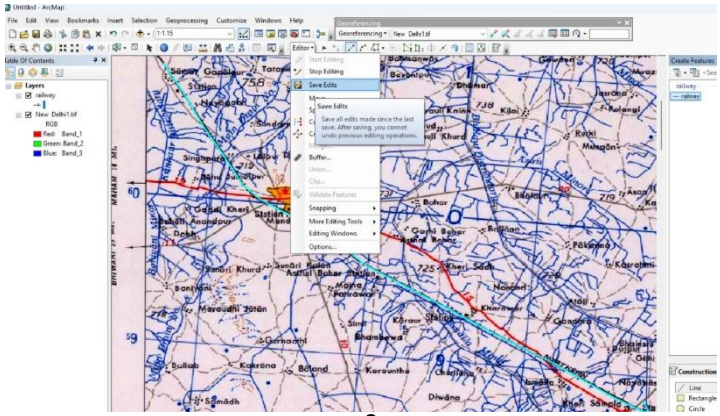


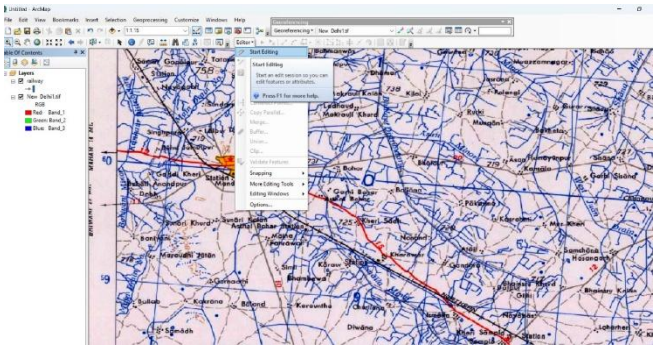
Step 2. Load Basemap or Imagery: Add a basemap such as satellite imagery or the toposheet itself to your map project to begin the digitizing process.

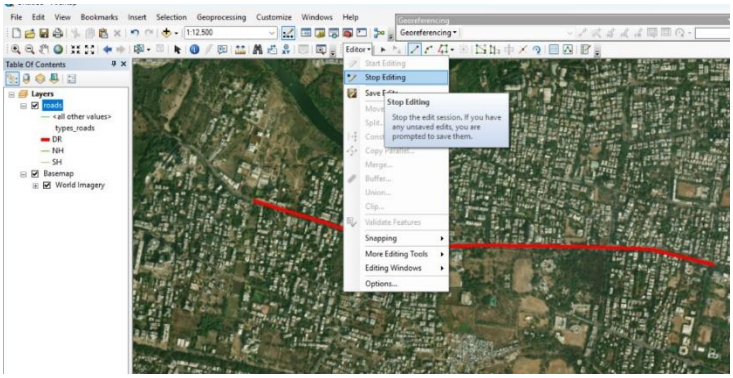


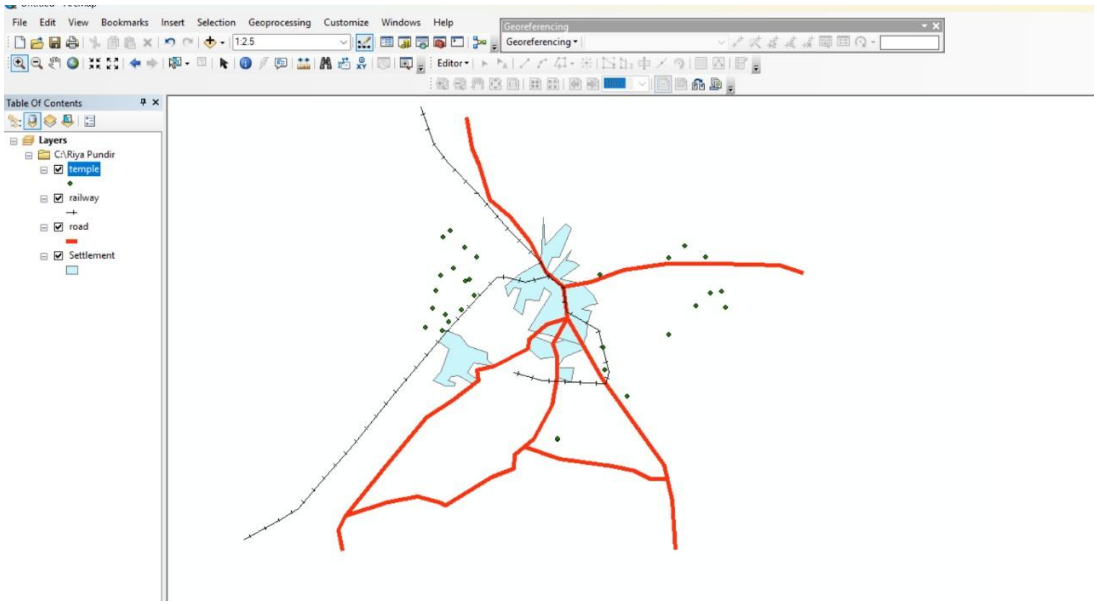
Serve as a reference for digitizing import the imagery use the Add Data tool to base map.

Step 3. Create or load a Feature class or shapefile: If you existing feature class, create one by right-clicking a geodata base in the catalog pane, Selecting New> Feature class and shapefile and defining its geometry type (point, line or polygon)









Digitizing points:

(1) Start Editing: open the Editor tool bar (Edit tab in Arc GIS pro or Editor menu in ARC Map). Select start Editing choosing the appropriate layer to edit

(2) Select Point Layer: In create Features pane, select the point layer you want to digitize.

(3) Digitize Points: Click on the map at the desired point features. Each click adds location to create a new point.

(4) Save Edits: Save your edits frequently by dicking Save edits.

Digitizing Lines:

(1) Start Editing & Select Line Layer: ensure you have selected the line feature class I shape tile in Create Features pane.

(2) Draw the Line: click on the starting point of the line, then click on subsequent points to create vertices along the path of the line. click or press Enter to Finish the line.

(3) Adjust Vertices (optional): use the Edit vertices tool to move, add, or delete vertices if needed.

(4) Save Edits: Save your edits to preserve the digitized lines.

Digitizing Polygons:

(1) Start Editing & Select Polygon Layer: In the create Features pane, choose the polygon rayes.

(2) Draw the Polygon: Click to place the first vertex of the polygon, then continue clicking to add more vertices.
Ensure you close the polygon by double-clicking on the starting point or pressing F2 to auto complete

(3) Modify Shape (optional): Use the Reshape Feature tool to adjust the polygon if necessary.

(4) Save Edits: Save your edits to secure your work.

Conclusion

The digitization of point, line, and polygon features in ArcMap helps convert real-world geographic information into accurate digital layers.Through this practical, we learned how to create and edit different feature classes, maintain topological accuracy, and prepare clean spatial data for further GIS analysis. This process forms the foundation for building reliable maps and geodatabases.

Practical 3: Working with Geodatabases in ArcMap

Aim

To create, manage, and organize spatial data using a **Geodatabase** in ArcMap for efficient storage, editing, and analysis.

Objectives

- To understand the structure and types of geodatabases.
- To learn how to create a **File Geodatabase** in ArcCatalog/ArcMap.
- To create feature classes (point, line, polygon) within a geodatabase.
- To import shapefiles and manage datasets within the geodatabase.
- To apply domains and subtypes for attribute consistency.

Introduction

A geodatabase is an organized system used to store and manage geographic information. It keeps different types of spatial data—such as feature classes, raster images, and tables together in one place. Because it works like a relational database, it helps ensure that the data remains accurate, consistent, and easy to maintain.

Key Features and Benefits

Centralized Data Storage: All spatial data is kept in one structured location, making management easier.

Better Data Accuracy: Built-in rules help maintain data quality and prevent errors.

Versioning: Multiple people can edit or work on the data at the same time without creating conflicts.

Advanced Spatial Analysis: Geodatabases support complex GIS analyses like terrain modelling, network analysis, and geostatistical tools.

Data Sharing & Collaboration: Data stored in a geodatabase can be accessed and used by many users, improving collaboration.

Types of Geodatabases

1. File Geodatabase: Stores geographic data in a single folder that contains multiple files. Best suited for small projects and individual users.

2. Personal Geodatabase: Uses a Microsoft Access database to store GIS data. It allows only one person to edit the data at a time.

3. Workgroup Geodatabase: Stores data using Microsoft SQL Server Express. It supports several users working on the data at the same time.

4. Enterprise Geodatabase: Uses large, full-featured database systems like Oracle, SQL Server, or PostgreSQL. It can handle big datasets and supports many users, making it ideal for large organizations.

Common Use Cases

Urban Planning: Studying land use, transportation systems, and population patterns.

Environmental Monitoring: Observing changes in climate, ecosystems, and natural resources.

Natural Resource Management: Managing forests, water, minerals, and other resources. Water resources and wildlife habitats can also be mapped and monitored using a geodatabase.

Public Safety: Used for studying crime patterns, planning emergency responses, and supporting disaster recovery.

Telecommunications: Helps in planning and managing network infrastructure.

Benefits of Using a Geodatabase

Improved Data Management: All spatial and attribute data is stored in one organized place, making it easier to manage.

Better Data Quality: Geodatabases include built-in tools for validation and editing, which help maintain clean and accurate data.

Advanced Spatial Analysis: Supports detailed and complex GIS analyses.

Collaboration: Allows multiple users to access and edit the same data at the same time.

Interoperability: Works well with different GIS software and data formats.
By using a geodatabase, organizations can manage and analyze spatial information more effectively, which leads to better planning and decision-making.

By using a geodatabase, organizations can effectively manage and analyze information leading to better decision making and problem solving.

File Geodatabase vs. Personal Geodatabase:

Both file geodatabases and personal geodatabases are used to store and manage GIS data, but they differ in structure and usage:

File Geodatabase:

Stores data in a folder with multiple files. It is faster, supports larger datasets, and allows multiple users to read the data.

Structure: Stored as a folder on disk, containing multiple files for different data types (feature classes, raster, tables)

Size limit: Can Store very large datasets, limited only by available disk space

Performance: Generally offers better performance, especially for large datasets.

Multi user editing: Supports multi ser editing, allowing multiple users to work on the same data simultaneously.

Data compression: Efficient data compression for smaller file sizes.

Security: Provides robust security mechanisms to protect dada.

Attachments: Can store attachments (e.g., Photos, documents) with features paraphrase it

Network Datasets: Supports network dataset for network analysis

Data versioning: Allows multiple versions of a dataset.

Personal Geodatabase:

Uses a single Microsoft Access (.mdb) file to store data. It has limited storage types of geodata bares used to store and manage geographic data in Are GIS. distinct characteristics and However, they have limited storage.

Structure: stored as a single microsoft Access file (@mdb)

Size limit: limited to 2GB in size.

Performance: may suffer performance issues with large datasets.

Multiuser Editing: single user editing only. Data compression; limited compression capabilities

Security: limited compression I searity capabilities

Attachments: Doesn't support attachments.

Network Datasets: Doesn't support network datasets.

Data versioning: Doesn't support versioning.

When to use which:

File Geodatabase:

- Large dataset
- Multiple users
- Advanced Spatial analysis and data management
- Complex data structures

Personal Geodatabase:

- Small, Single user projects
- Basic spatial analysis
- Limited data storge needs

Data Used: We have taken the toposheet of Surat, Gujarat, downloaded from Survey of India.

Toposheet no: G43H3 (45F/3)

Scale 1:250,000

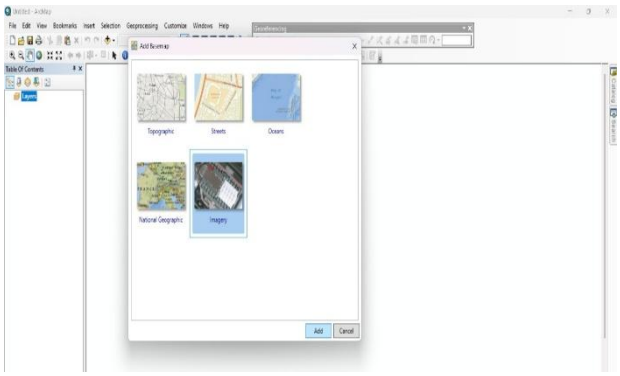
Methodology

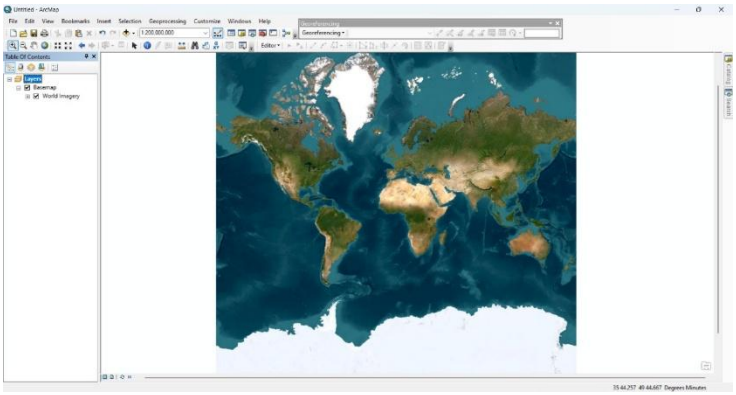
CREATION OF GEAODATABASE

Step 1: Go to Catalog and click on the folder where we want to create the Geodatabase.

Step 2: then click "New" > "personal database"

Step 3: Then the personal Geodatabase will be created.

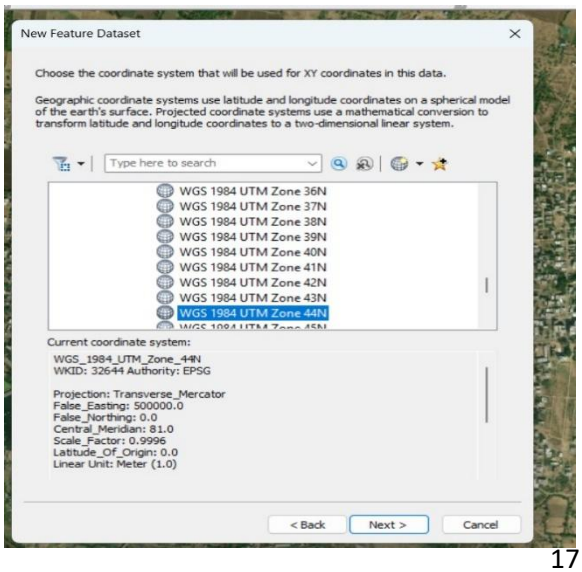


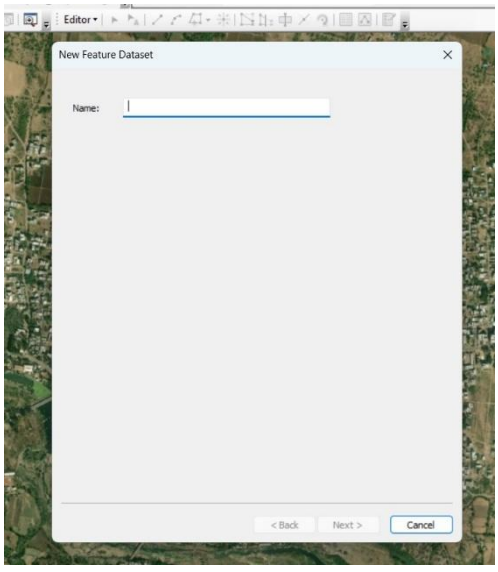


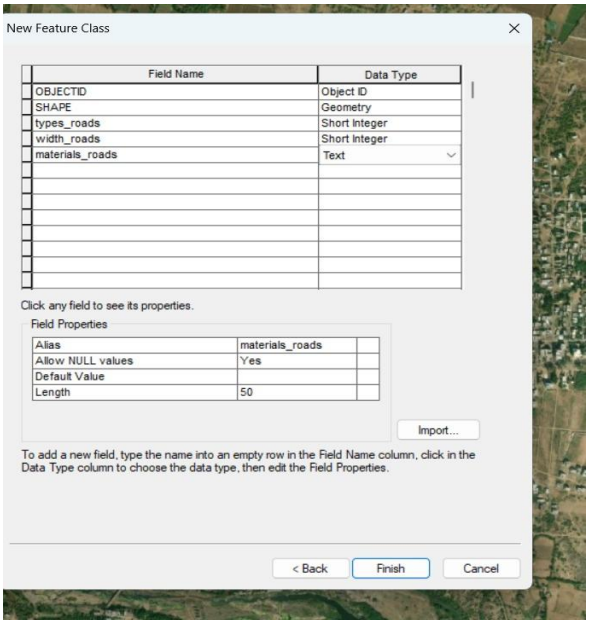
CREATING THE FEATURE DATASET

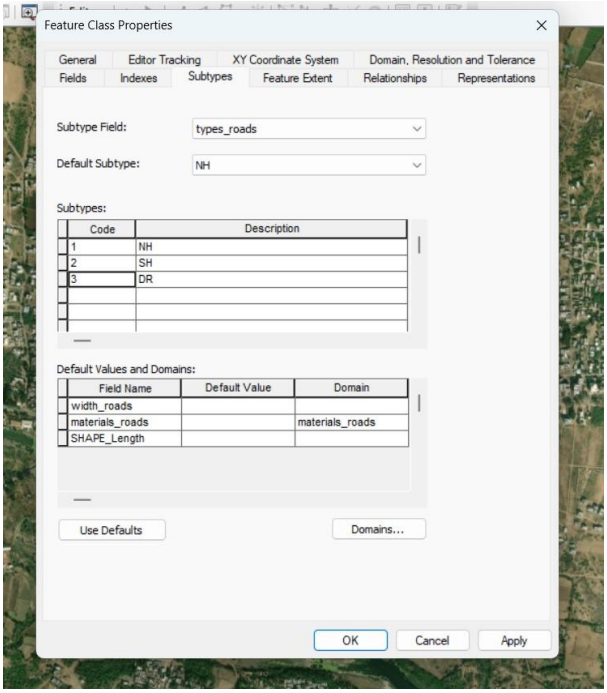
Step 1: Then Click on the existing Geodatabase and. sight click New> the feature dataset.

Step 2: Give a name select the coordinate system. and finally click on the 'Finish'







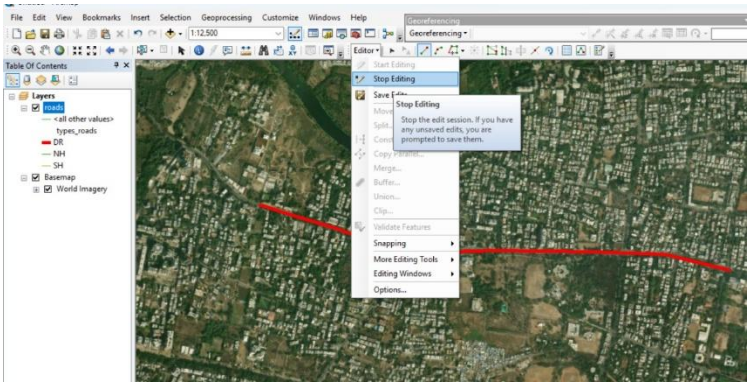


CREATING FEATURE CLASS & SUB TYPES:

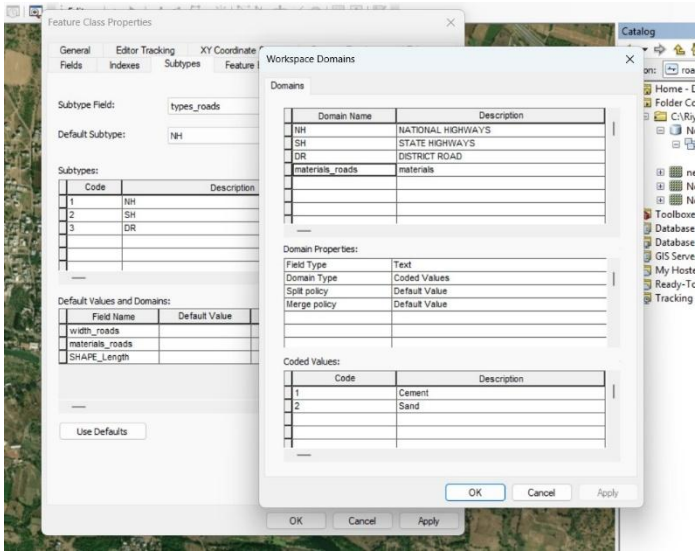
Step 1: Click on the existing feature dataset > feature class after clicking new.

Step2: Write the name and choose the type of Features stored in this feature class, if we take the polyline feature it shows and digitizes the line feature

Step 3: After clicking next there will be a window opened where we have to select and write down the Sub types which we want.



Steps 4: After that we click on ok or Finish and then the Feature class will be added.



CRATING DOMAIN

Step 1: After creating the feature dataset, go to the properties and there the subtype option will be open.

Step 2: There we have to write down the code and description in the subtype section.

Step 3: In the corner domain option is there, click on this, and then the domain window will open.

Step 4: Then we have to fill the domain section & create the domains and then after completing all click on the apply and lastly ok.

Steps 5: In the domain section we have to select the code value and range according to our need.

Step 6: After that, we can start digitizing the existing map.

Practical 4: Scale

scale represents the ratio of a distance on a map to the corresponding distance on the ground, allowing users to understand the relationship between the two and accurately measure features. It is crucial for understanding map proportions and is expressed in three main ways: as a representative fraction (e.g., 1:24,000), a verbal statement (e.g., "1 inch equals 1 mile"), or a linear or bar scale (a graphical representation)

the ratio between the distance on map and the actual distance on the ground." This can be expressed as: The representation of earth or even a piece of area cannot be done on a piece of paper practically without fixing a proportional geometric shape or symbols representing the features on the earth as paper cannot always be of the same size of the area under representation. For this reason, the maps are drawn on small piece of papers of varying sizes using some representative proportions where certain distance or size of representing features on the paper is true to the fixed proportionate size on the ground. For example, one centimetre line on paper may represent one kilometre distance on the actual ground.

- **Large-scale maps** show a smaller geographic area in greater detail. They have a relatively large representative fraction, such as **1:10,000** or 1:25,000, meaning the area on the map is closer to its real-world size. These are used for city plans, hiking maps, and property surveys.
- **Small-scale maps** show a large geographic area with less detail. They have a relatively small representative fraction, such as **1:1,000,000** or 1:10,000,000, and are used for world maps or maps of continents

REPRESENTATIVE METHODS OF SCALE

After defining the scale, one of the obvious questions that come to our mind is about the methods of showing scale on a map.. These are as follows:

1. Statement of Scale
2. Linear or Graphic Scale
3.Representative Fraction (R.F.)

- **Representative Fraction (RF) or Ratio Scale:** This scale is expressed as a unitless ratio or fraction, such as **1:100,000** or **1/100,000**. This means that one unit of any measurement on the map (e.g., 1 centimeter, 1 inch) represents 100,000 of those same units in the real world. This method is considered universal because it is independent of specific units of measurement.

- **Verbal or Statement Scale:** This method uses words to describe the relationship between map distance and real-world distance. For example: "**1 inch represents 1 mile**" or "**1 centimeter to 10 kilometers**". This is easy to understand but can be problematic if the map is used by someone unfamiliar with the specified units, or if the map is photographically enlarged or reduced, which would render the written statement inaccurate.

- **Linear or Bar Scale:** This is a visual representation using a straight line or bar marked with segments that represent actual distances on the ground (e.g., kilometers, miles). A key advantage is that if the map is enlarged or reduced, the bar scale changes size proportionally and remains accurate, unlike a statement scale.

TYPES OF MAP SCALE : CONSTRUCTION AND READING

PLAIN SCALE

Plain scales are those which measure up to two units or a single unit through its divisions and sub-divisions. Let us explain this with examples. Example of two units of measurement is 2 kilometer and 200 meter. Examples of divisions and sub-divisions are 5.4 cm and 3.8 meter, etc.

plain scale represents up to two units, therefore, it consists of a line divided into number of equal main parts and the first main part is sub-divided into smaller parts. Zero (0), is marked at the end of the first main part. From zero mark numbers to the main parts or units towards right and give numbers to the sub-divisions or smaller parts towards left. Give the names of the units and sub-units below clearly.

COMPARITIVE SCALE

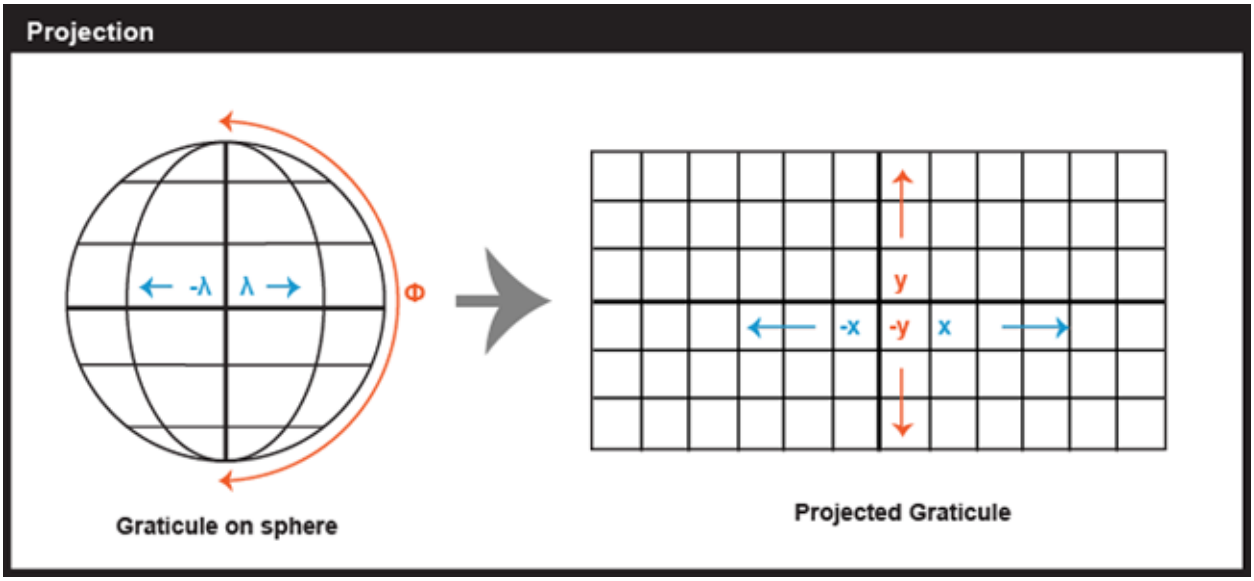
In simpler terms, a comparative scale may be expressed as a pair of scales having a common R.F. but graduated to read different units. Some of the examples of comparative scales are (i) different units; (ii) time scale; (iii) pace scale; and (iv) revolution scale

DIAGONAL SCALE

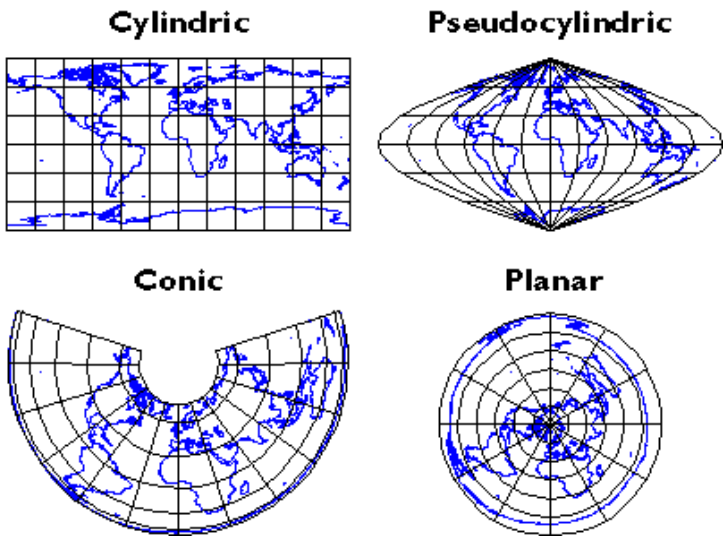
Diagonal scales are drawn for greater precision or higher degree of accuracy. Can you imagine how is this precision achieved? Yes, you're right. We achieve this precision because this scale is specifically used to measure up to three units. For example, in a metric system you can measure kilometers (km), meters (m), and centimeters (cm). Similarly, in British system of measurement you can measure miles, furlongs and yards etc. If you observe closely the previous two scales, they are best suited for measuring up to two units. This scale is used when very small distances such as 0.1 mm are to be accurately measured or when measurements are required up to second decimal.

Practical 5: Projection: (Conical& Meracator)

Latitude and longitude coordinates specify positions in a spherical grid called the **graticule** (that approximates the more-or-less spherical Earth). The true geographic coordinates called **unprojected** coordinate in contrast to plane coordinates, like the Universal Transverse Mercator (UTM) and State Plane Coordinates (SPC) systems, that denote positions in flattened grids. These georeferenced plane coordinates are referred to as ***projected***. The mathematical equations used to project latitude and longitude coordinates to plane coordinates are called map projections. Inverse projection formulae transform plane coordinates to geographic. The simplest kind of projection, illustrated below, transforms the graticule into a rectangular grid in which all grid lines are straight, intersect at right angles, and are equally spaced. Projections that are more complex yield grids in which the lengths, shapes, and spacing of the grid lines vary. Even this simplest projection produces various kinds of distortions; thus, it is necessary to have multiple types of projections to avoid specific types of distortions. Imagine the kinds of distortion that would be needed if you sliced open a soccer ball and tried to force it to be completely flat and rectangular with no overlapping sections. That is the amount of distortion we have in the simple projection below (one of the more common in web maps of the world today).



There are three main categories of map projection, those in which projection is directly onto a flat plane, those onto a cone sitting on the sphere that can be unwrapped, and other onto a cylinder around the sphere that can be unrolled (Figure 2.15 above). All three are shown in their normal aspects. The plane often is centered upon a pole. The cone is typically aligned with the globe such that its line of contact (tangency) coincides with a parallel in the mid-latitudes. Moreover, the cylinder is frequently positioned tangent to the equator (unless it is rotated 90°, as it is in the Transverse Mercator projection). As you might imagine, the appearance of the projected grid will change quite a lot depending on the type of surface it is projected onto, how that surface is aligned with the globe, and where that imagined light is held. The following illustrations show some of the projected graticules produced by projection equations in each category.



- **Cylindric projection** equations yield projected graticules with straight meridians and parallels that intersect at right angles. The example shown above is a Cylindrical Equidistant (also called Plate Carrée or geographic) in its normal equatorial aspect.
- **Pseudocylindric projections** are variants on cylindrics in which meridians are curved. The result of a Sinusoidal projection is shown above.
- **Conic projections** yield straight meridians that converge toward a single point at the poles, parallels that form concentric arcs. The example shown above is the result of an Albers Conic Equal Area, which is frequently used for thematic mapping of mid-latitude regions.

- **Planar projections** also yield meridians that are straight and convergent, but parallels form concentric circles rather than arcs. Planar projections are also called **azimuthal** because every planar projection preserves the property of azimuthality, directions (azimuths) from one or two points to all other points on the map. The projected graticule shown above is the result of an Azimuthal Equidistant projection in its normal polar aspect.

Map Projections: Distortion

No projection allows us to flatten the globe without distorting it. Distortion ellipses help us to visualize what type of distortion a map projection has caused, how much distortion occurred, and where it occurred. The ellipses show how imaginary circles on the globe are deformed because of a particular projection. If no distortion had occurred in the process of projecting the map shown below, all of the ellipses would be the same size, and circular in shape.

When positions on the graticule are transformed to positions on a projected grid, four types of distortion can occur: distortion of sizes, angles, distances, and directions. Map projections that avoid one or more of these types of distortion are said to preserve certain properties of the globe: equivalence, conformality, equidistance, and azimuthality, respectively

Equivalence

equal-area projections maintain correct proportions in the sizes of areas on the globe and corresponding areas on the projected grid (allowing for differences in scale, of course). Notice that the shapes of the ellipses in the Cylindrical Equal Area projection above are distorted, but the areas each one occupies are equivalent. Equal-area projections are preferred for small-scale thematic mapping (discussed in the next chapter), especially when map viewers are expected to compare sizes of area features like countries and continents.

Conformality

The distortion ellipses plotted on the **conformal** projection shown above vary substantially in size, but are all the same circular shape. The consistent shapes indicate that conformal projections preserve the fidelity of angle measurements from the globe to the plane. In other words, an angle

measured by a land surveyor anywhere on the Earth's surface can be plotted at its corresponding location on a conformal projection without distortion. This useful property accounts for the fact that conformal projections are almost always used as the basis for large scale surveying and mapping. Among the most widely used conformal projections are the Transverse Mercator, Lambert Conformal Conic, and Polar Stereographic.

Equidistance

Equidistant map projections allow distances to be measured accurately along straight lines radiating from one or, at most, two points or they can have correct distance (thus maintain scale) along one or more lines. In the example below (also sometimes called an "equirectangular" projection because the parallels and meridians are both equally spaced). Notice that ellipses plotted on the Cylindrical Equidistant (Plate Carrée) projection shown above vary in both shape and size. The north-south axis of every ellipse is the same length, however. This shows that distances are true-to-scale along every meridian; in other words, the property of equidistance on this map projection is preserved from the two poles.

Azimuthality

Azimuthal projections preserve directions (azimuths) from one or two points to all other points on the map. **Gnomonic projections**, like the one above, display all great circles as straight lines. A **great circle** is the most direct path between two locations across the surface of the globe. See how the ellipses plotted on the gnomonic projection shown above vary in both size and shape, but are all oriented toward the center of the projection. In this example, that is the one point at which directions measured on the globe are not distorted on the projected graticule. This is a good projection for uses like plotting airline connections from one airport to all others.

Compromise

Some map projections preserve none of the properties described above, but instead seek a compromise that minimizes distortion of all kinds. The example shown above is the Polyconic

projection, where parallels are all non-concentric circular arcs, except for a straight equator, and the centers of these circles lie along a central axis. The U.S. Geological Survey used the polyconic projection for many years as the basis of its topographic quadrangle map series until the conformal Transverse Mercator succeeded it. Another example is the Robinson projection, which is often used for small-scale thematic maps of the world (it was used as the primary world map projection by the National Geographic Society from 1988-1997, then replaced with another compromise projection,

Practical 6: Interpretation of Maps

CONTOUR

INTERPRETATION

This map shows contour lines, which help us understand the shape and height of the land. Even though some of the numbers on the lines are unclear, we can still figure out the basic features of the terrain.

(i)TypeofTerrain

The contour lines are uneven and spread out in different directions, which suggests the area is hilly or mountainous. The height changes are not smooth, and there may be ridges, valleys, and natural drainage paths.

(ii) Steep vs. Gentle Slopes

- When contour lines are very close to each other, the slope is steep. These steep areas can be seen in many parts of the map, especially in the middle and toward the upper-right side.
- When contour lines are spaced far apart, the slope, the slope is gentle contour li These gentler slopes appear more in the lower-central region of the map.

(iii)Valleys/WaterFlow

If the contour lines form a V-shape that points uphill, it usually means there is a valley or a stream flowing through that area.

VEGETATION

INTERPRETATION

The tracing sheet displays irregularly shaped patches of varying sizes, which generally indicate different vegetation types or categories of land cover.

(i) Large, elongated patches – Forested or dense vegetation zones

- The long, narrow configurations suggest areas of continuous forest cover.
- These forms may also represent woodland distributed along slopes or ridge lines, reflecting vegetation patterns influenced by natural landforms such as hills and valleys.

(ii) Broad, irregular blocks – Mixed or plateau vegetation

- The larger, irregular ("blob-like") shapes observed toward the upper and central portions likely correspond to:
    - Mixed woodland communities
    - Tall grasslands interspersed with tree cover

(iii) Small central patches

- The smaller patches near the center typically signify:
    - Exposed or bare rock surfaces
    - Localized shrub clusters
    - Isolated individual trees

CULTURAL FEATURES

INTERPRETATION

The tracing sheet highlights various man-made features using specific lines and symbols.

(i) Transportation Infrastructure

- **Roads:** Thick, solid continuous lines indicate primary roads or major tracks.
- **Paths or Unmetalled Roads:** Dashed or dotted lines likely represent footpaths, tracks, bridleways, or unpaved secondary roads.
- **Junctions/Intersections:** Points where a solid road meets or intersects with a dashed/dotted line suggest the presence of a road junction.

(ii) Settlement Patterns

- **Settlement/Building Cluster:** Small, irregular, enclosed shapes generally signify a village, hamlet, or a cluster of isolated buildings or farmsteads.

(iii) Possible Natural/Mixed Features

- **Water Channels:** Some thinner, wavy lines may denote streams, drainage lines, or irrigation channels.

Practical 7: Cross Section Profile

A cross-section profile is a **2D graph that shows the vertical shape of the land or an object as if it were sliced**. It is created by drawing a line across a map or 3D model and then plotting the elevation at each point along that line, creating a side-view perspective that reveals features like hills, valleys, or internal structures. This is a useful tool in fields like surveying, geology, and engineering for visualizing and analyzing terrain and designs for projects like roads or canals.

Key characteristics:

- **Graphical representation:** It is a graph with elevation on the y-axis and distance along the section line on the x-axis.
- **Side-view perspective:** It provides a view of the surface as if you were looking at a slice through the terrain or object.
- **Uses:**
    - **Surveying and civil engineering:** To visualize and design infrastructure like roads, railways, and canals by showing the ground's gradient and features.
    - **Geology:** To show the internal structure of the Earth by representing rock layers and features.
    - **Technical drawing:** To show the internal arrangement of a 3D object on a 2D plane.

How it's made:

1. **Draw a section line:** Choose and draw a straight line across a map or 3D model to represent the path of the cross-section.
2. **Determine elevations:** Note the elevation at points along the line, particularly where contour lines intersect it.
3. **Plot the profile:** Create a new graph. Plot the points from the map onto the graph, using the distance along the line as the x-coordinate and the corresponding elevation as the y-coordinate.
4. **Connect the points:** Draw a line connecting the plotted points to create the final cross-section profile.

Common applications:

- **Engineering projects:** Calculating volumes, designing earthworks, and analyzing stability for projects like highways and dams.
- **Topographic mapping:** Converting 2D contour maps into a more easily understandable 3D side-view profile.
- **Geological studies:** Analyzing the subsurface structure of an area to understand its geology and potential resources